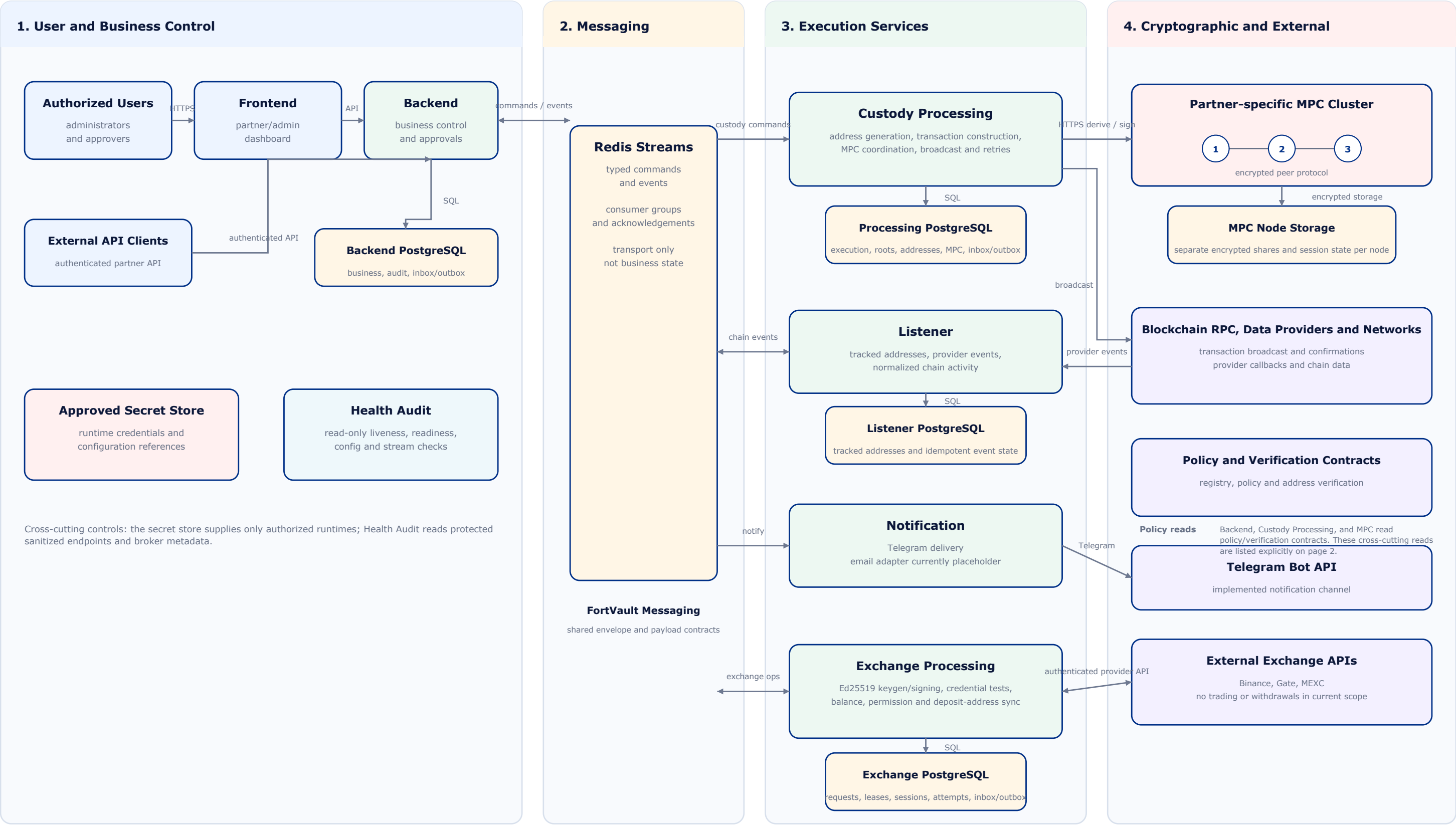


FortVault Runtime Architecture

Components, owned state, runtime connections, and trust boundaries

User/client Service Owned durable state Cryptographic/secret bound External/on-chain Operational control



FortVault Connection Register

Authoritative summary of the runtime connections shown on page 1

Source	Destination	Interface	Purpose	Status
Authorized user	Frontend	HTTPS	Custody administration, review and approval	Current
Frontend	Backend	Authenticated HTTPS API	Partner-scoped business operations	Current
External API client	Backend	Authenticated HTTPS API	Partner API integration where enabled	Current
Backend	Backend PostgreSQL	PostgreSQL	Business, audit, inbox and outbox state	Current
Backend	Redis Streams	Typed Messaging envelopes	Publish commands and consume completion events	Current
Redis Streams	Custody Processing	Consumer group	Address and transfer execution commands	Current
Custody Processing	Redis Streams	Durable outbox	Address and transfer completion events	Current
Custody Processing	Processing PostgreSQL	PostgreSQL	Execution, retries, roots, addresses and MPC state	Current
Custody Processing	MPC master node	Authenticated HTTP API	Bootstrap keygen, derivation and exact-payload signing	Current
MPC node	MPC peers	Encrypted gRPC	Interactive threshold cryptographic protocol	Current
MPC node	Own node storage	PostgreSQL/local boundary	Encrypted key share and MPC session state	Current
Backend	Policy contracts	Blockchain read calls	Read partner policy and verification configuration	Current
Custody Processing	Policy contracts	Blockchain read calls	Validate wallet types, policy and execution context	Current
MPC cluster	Policy contracts	Blockchain read calls	Independent authorization and policy validation	Current
Custody Processing	RPC / blockchain	Chain-specific RPC	Build, broadcast and query custody transactions	Current
Chain data provider	Listener	Webhook / polling API	Chain activity and confirmation data	Current
Listener	Listener PostgreSQL	PostgreSQL	Tracked addresses and idempotent event state	Current
Listener	Redis Streams	Typed normalized events	Deposits, confirmations and transaction status	Current
Redis Streams	Notification	Consumer group	Notification commands	Current
Notification	Telegram Bot API	HTTPS provider API	Configured Telegram delivery	Current
Notification	Email adapter	Internal adapter	Logs intended email; provider not yet integrated	Placeholder
Redis Streams	Exchange Processing	Consumer group	Exchange operation commands	Current
Exchange Processing	Exchange PostgreSQL	PostgreSQL	Requests, leases, sessions, attempts and outbox	Current
Exchange Processing	MPC master node	Authenticated HTTP API	Ed25519 key generation and exact query signing	Current
Exchange Processing	Exchange APIs	Authenticated HTTPS API	Credential test, balance, permission and deposit sync	Current
Exchange Processing	Redis Streams	Durable outbox	Sanitized exchange completion events	Current
Approved secret store	Authorized services	Secret-management API	Runtime configuration and credential material	Current
Health Audit	Service / MPC health	Read-only HTTPS	Liveness, readiness and sanitized configuration	Current
Health Audit	Redis metadata	Read-only Redis commands	Expected streams and consumer groups	Current

Development prototypes outside the production runtime boundary

Bootstrap Station and Cold Signer implement development work toward offline ECDSA MPC bootstrap/signing. They remain non-production prototypes; the intended architecture, security invariants, and remaining readiness work are documented in architecture/offline-mpc.md.